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Advanced Topics in Econometrics
(PhD in Mathematics Applied to Economics and Management)
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Exercise Sheet 10 - Panel Data Models
(version 4/11/2025)

1. Consider the Fixed Effects Model:

$$\begin{aligned} y_{it} &= \alpha_i + x_{it}'\beta + \varepsilon_{it}, \\ i &= 1,...,n \\ t &= 1,...,T, \end{aligned} \tag{1}$$

where $\varepsilon_{it} \sim IID\left(0, \sigma_{\varepsilon}^2\right)$ and x_{it} is $(k-1)\times 1$ vector of explanatory variables <u>not</u> including a constant and including only regressors that vary with t, where ε_{it} is independent of x_{it} for all i and t and α_i is potentially correlated with x_{it}.

(a) Show that the "within estimator"

$$\begin{aligned} \hat{\beta}_{within} &= \left[\sum\nolimits_{i=1}^{n}\sum\nolimits_{t=1}^{T}\left(x_{it}-\bar{x}_i\right)\left(x_{it}-\bar{x}_i\right)'\right]^{-1} \\ &\times\left[\sum\nolimits_{i=1}^{n}\sum\nolimits_{t=1}^{T}(x_{it}-\overline{x}_i)(y_{it}-\overline{y}_i)\right] \end{aligned}$$

is unbiased estimator of β, where $\overline{x}_i = T^{-1}\sum_{t=1}^{T} x_{it}$ and $\overline{y}_i = T^{-1}\sum_{t=1}^{T} y_{it}$.

(b) Use the Frisch–Waugh theorem to show that the "least squares dummy variable" (LSDV) estimator for β is identical to the within estimator.[1]

[1] **Frisch-Waugh theorem:** Consider the partitioned linear regression model $Y = X_1\beta_1 + X_2\beta_2 + U$, where Y is $(n\times 1)$, β_1 is $(k_1\times 1)$ and β_2 is $(k_2\times 1)$, $k_1+k_2=k$; X_1 is $(n\times k_1)$ and X_2 is $(n\times k_2)$, $rank(\ (X_1,X_2))=k$. Let $M_1 = I_n - X_1(X_1'X_1)^{-1}X_1'$, then $\hat{\beta}_2 = (X_2'M_1X_2)^{-1}X_2'M_1Y$.